

## CST8912 – Cloud Solution Architecture

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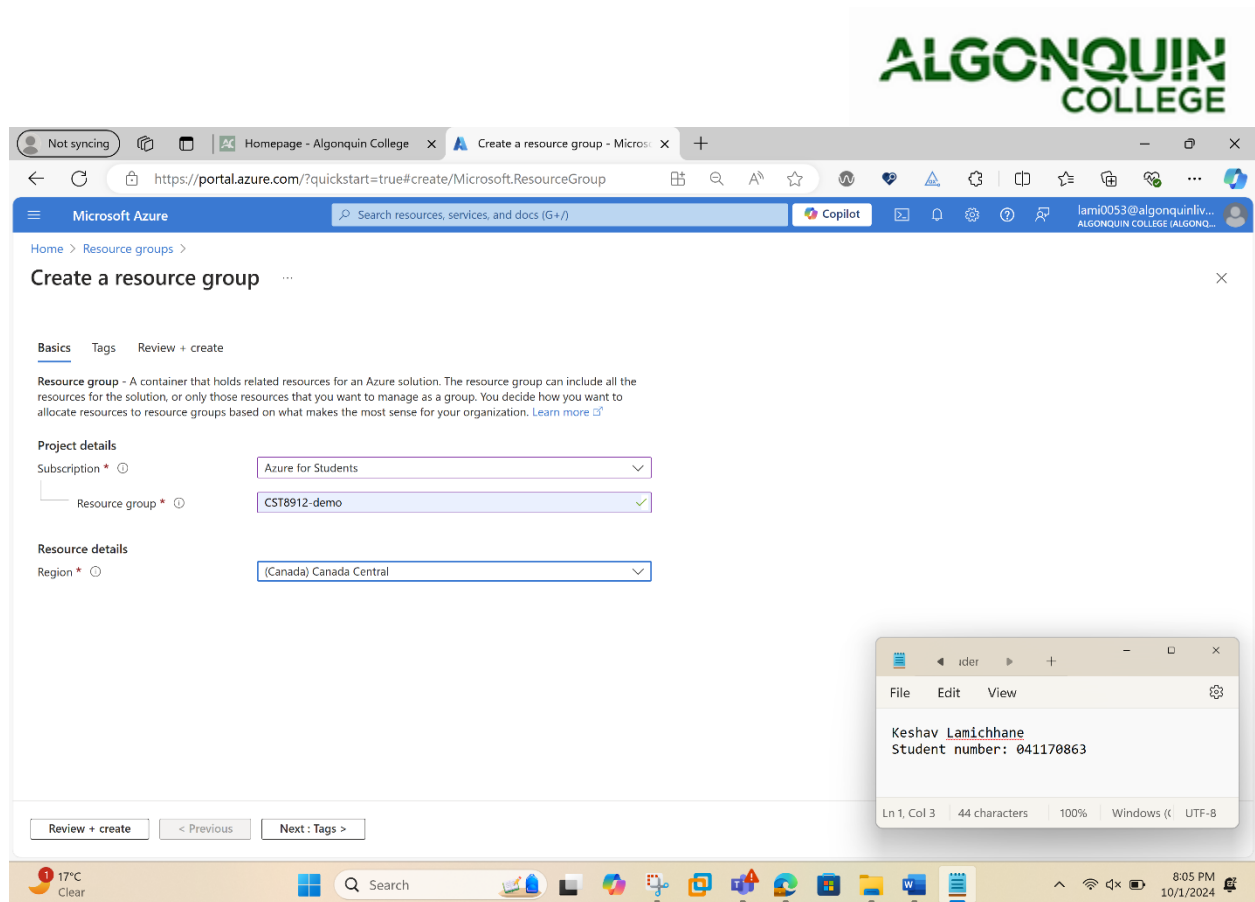
### Graded Lab Activity #4

**Introduction:** A cloud database is a type of database service that utilizes the flexibility of cloud computing to provide features like traditional databases, all while being developed and accessed through a cloud platform. By eliminating the requirement for specialized hardware, it enables businesses to host databases and drastically lowers infrastructure expenses. Businesses have the option of choosing between more control and easier management using cloud databases, which can be handled directly by users or provided as a managed service by cloud providers. They support a wide range of database types, including NoSQL databases like MongoDB and Apache CouchDB and traditional databases like MySQL and PostgreSQL. These databases are usually accessible through vendor-provided APIs or web interfaces, which improves scalability and accessibility in contemporary cloud systems.

The primary functions of the cloud SQL database for Azure will be covered in this lab. We will also learn how to connect to the database, query it, and use Azure Data Factory to copy data from one location to another, as well as deploy and maintain the service. Azure Storage Account Container is the destination in this lab, while Azure SQL Database is the source, which is a Microsoft PaaS service.

### Procedures:

Step 1: Set up an Azure SQL database in the resource group cst8912-demo located in the Canada Central region. In the SQL deployment configuration, under SQL databases, choose the single database option.



Step 2: On the create database page, enter the following values, leaving the remaining attributes at their default settings.

Membership: Choose your Azure plan.

Group of resources: CST8912-demo

Name of database: db8912

Home > SQL databases >

## Create SQL Database

Microsoft  
manage all your resources

Subscription \*

Resource group \*   
[Create new](#)

### Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name \*

Server \*   
[Create new](#)

Want to use SQL elastic pool? ☐ Yes ☒ No

Workload environment  
☒ Development  
☐ Production

Default settings provided for Development workloads. Configurations can be modified as needed.

[Review + create](#) [Next: Networking >](#)

Keshav Lamichhane  
Student number: 041170863

In this step we need to create a new server in Canada central region. And select sql authentication and mention the name for server admin and password too.

Server : db8912demo

Username:db8912\_keshav

Password: anything

Want to use SQL elastic pool? No

Workload environment: Development

Compute + storage: unchanged

Backup storage redundancy: Locally-redundant backup storage

Not syncing | Homepage - Algonquin College | Create SQL Database Server - Microsoft | New tab

https://portal.azure.com/?quickstart=true#view/SqlAzureExtension/Shared...

Microsoft Azure | Search resources, services, and docs (G+/)

Home > SQL databases > Create SQL Database >

## Create SQL Database Server

Microsoft

Location \* (Canada) Canada Central

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login \* db8912\_keshav

Password \*

Confirm password \*

OK

17°C Clear | Search | 8:14 PM 10/1/2024

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File Edit View

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Not syncing | Homepage - Algonquin College | Create SQL Database - Microsoft | New tab

https://portal.azure.com/?quickstart=true#create/Microsoft.SQLDatabase

Microsoft Azure | Search resources, services, and docs (G+/)

Home > SQL databases >

## Create SQL Database

Microsoft

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name \* db8912

Server \* (new) db8912demokeshav (Canada Central)

Want to use SQL elastic pool? ☐ Yes ☒ No

Workload environment

- ☒ Development
- ☐ Production

Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage \*

**General Purpose - Serverless**  
Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled  
[Configure database](#)

Backup storage redundancy

Review + create | Next: Networking >

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File Edit View

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Student number: 041170863

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Home > SQL databases >

## Create SQL Database

Microsoft

Want to use SQL elastic pool? ☐ Yes ☒ No

Workload environment ☒ Development ☐ Production

**Default settings provided for Development workloads. Configurations can be modified as needed.**

Compute + storage ☐ **General Purpose - Serverless**  
Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled  
[Configure database](#)

**Backup storage redundancy**

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy ☒ Locally-redundant backup storage ☐ Zone-redundant backup storage ☐ Geo-redundant backup storage

[Review + create](#) [Next: Networking >](#)

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Step 3: Choose Next:Networking > on the Create SQL Database page, and Public endpoint under Network connectivity on the Networking page. Then, to grant access to your database server from Azure services and your current client IP address, select Yes for both options in the Firewall rules section.



Not syncing

Homepage - Algonquin College

Create SQL Database - Microsoft

New tab

https://portal.azure.com/?quickstart=true#create/Microsoft.SQLDatabase

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lami0053@algonquinliv...  
ALGONQUIN COLLEGE (ALGONQ...

Home > SQL databases >

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'db8912demokeshav' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method

☐ No access

☒ Public endpoint

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server \*

☐ No ☒ Yes

Add current client IP address \*

☐ No ☒ Yes

Review + create < Previous Next: Security >



Cost summary

General Purpose (GP\_S\_Gen5\_1)

Cost per GB (in USD) 0.13

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH 5.26 USD

COMPUTE COST / VCORE SECOND 0.000174 USD

NOTES

1 Serverless databases are billed in vCore seconds on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

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File Edit View

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 Search



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10/1/2024

Step 4: In this step, select Next: Security > and set the Enable Microsoft Defender for SQL option to Not now.

Not syncing | Homepage - Algonquin College | Create SQL Database - Microsoft | New tab

https://portal.azure.com/?quickstart=true#create/Microsoft.SQLDatabase

Microsoft Azure | Search resources, services, and docs (G+)

Home > SQL databases >

## Create SQL Database

Microsoft

Basics Networking **Security** Additional settings Tags Review + create

**Microsoft Defender for SQL**

Protect your data using Microsoft Defender for SQL, a unified security package including vulnerability assessment and advanced threat protection for your server. [Learn more](#)

Get started with a 30 day free trial period, and then 15 USD/server/month.

Enable Microsoft Defender for SQL \* ☐ Start free trial ☒ Not now

**Ledger**

Ledger cryptographically verifies the integrity of your data and detects any tampering that might have occurred. [Learn more](#)

Ledger **Not configured** [Configure ledger](#)

**Server identity**

Use system assigned and user assigned managed identities to enable central access management between this database and other Azure resources. [Learn more](#)

Server identity **Not enabled**

[Review + create](#) < Previous Next : Additional settings >

Cost summary

**General Purpose (GP\_S\_Gen5\_1)**

Cost per GB (in USD) 0.13

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH 5.26 USD

COMPUTE COST / VCORE SECOND <sup>1</sup> 0.000174 USD

**NOTES**

<sup>1</sup> Serverless databases are billed in vCore seconds on a combination of CPU and memory utilization. [more about serverless billing](#)

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17°C Clear | Search | 8:16 PM 10/1/2024

Step 5: In this step click on additional setting and set the Use existing data option to Sample.

Not syncing | Homepage - Algonquin College | Create SQL Database - Microsoft | New tab

https://portal.azure.com/?quickstart=true#create/Microsoft.SQLDatabase

Microsoft Azure | Search resources, services, and docs (G+)

Home > SQL databases >

## Create SQL Database

Microsoft

Basics Networking Security **Additional settings** Tags Review + create

Customize additional configuration parameters including collation & sample data.

**Data source**

Start with a blank database, restore from a backup or select sample data to populate your new database.

Use existing data \* ☐ None ☐ Backup ☒ Sample

AdventureWorksLT will be created as the sample database.

**Database collation**

Database collation defines the rules that sort and compare data, and cannot be changed after database creation. The default database collation is SQL\_Latin1\_General\_CP1\_CI\_AS. [Learn more](#)

Collation

[Review + create](#) < Previous Next : Tags >

Cost summary

**General Purpose (GP\_S\_Gen5\_1)**

Cost per GB (in USD) 0.13

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH 5.26 USD

COMPUTE COST / VCORE SECOND <sup>1</sup> 0.000174 USD

**NOTES**

<sup>1</sup> Serverless databases are billed in vCore seconds on a combination of CPU and memory utilization. [more about serverless billing](#)

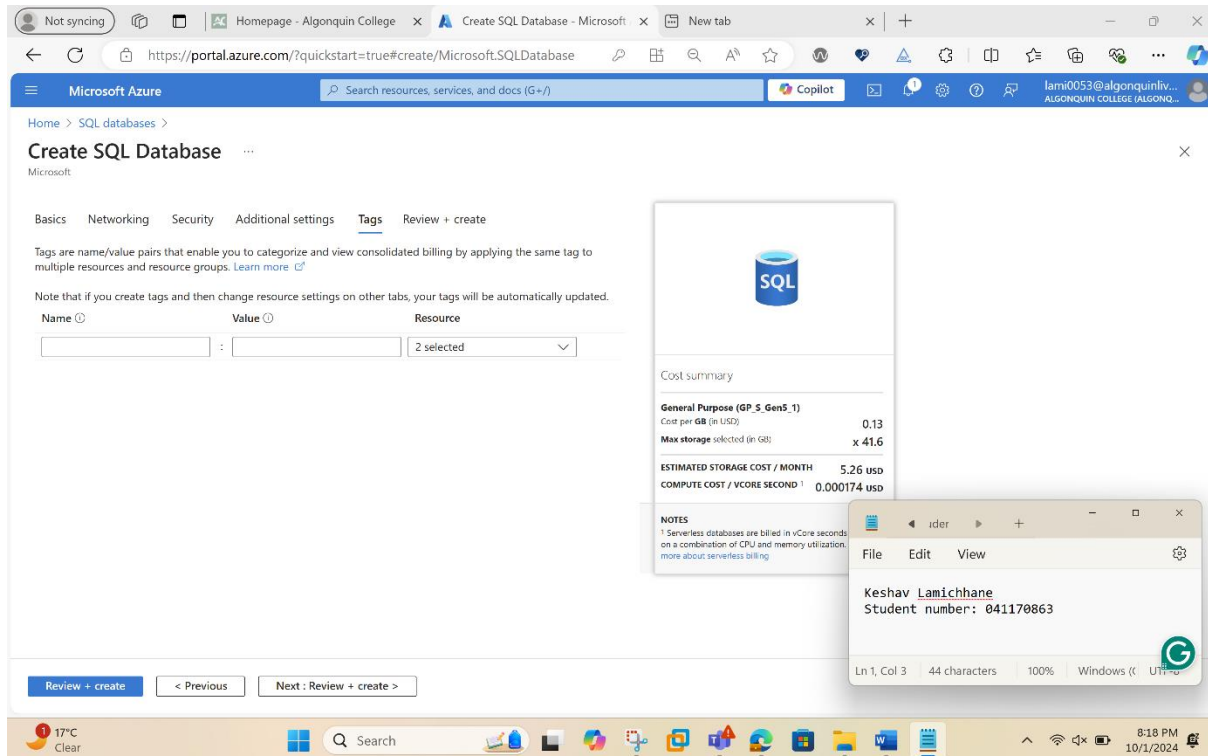
Keshav Lamichhane  
Student number: 041170863

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17°C Clear | Search | 8:18 PM 10/1/2024

Then follow following steps:

Step 6: Select Review + Create, and then select Create to create your Azure SQL database.



The screenshot shows the Microsoft Azure portal interface for creating a new SQL database. The 'Review + create' tab is active, displaying a cost summary and a 'Review + create' button. The cost summary indicates a general purpose (GP\_5\_Gen5\_1) database with a cost per GB of 0.13 USD, a max storage of 41.6 GB, and an estimated storage cost of 5.26 USD per month. The compute cost is 0.000174 USD per vCore second. The 'Review + create' button is highlighted in blue.

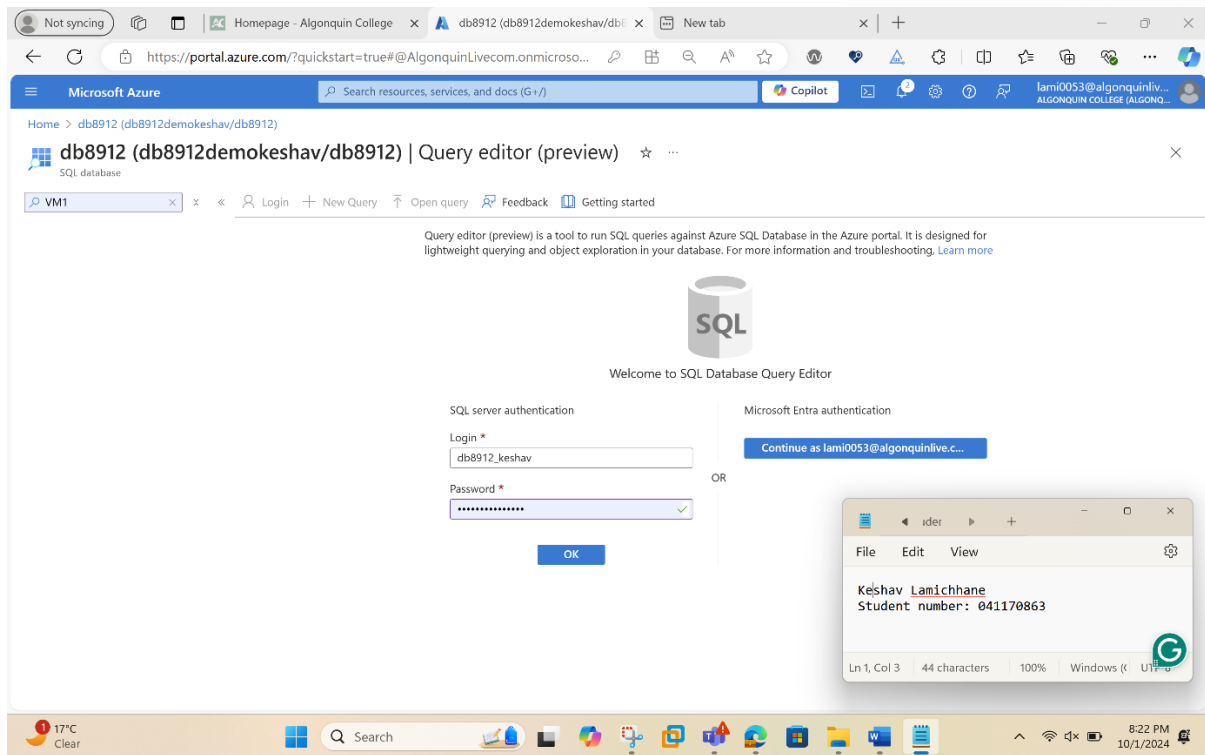
| Cost summary                   | Value        |
|--------------------------------|--------------|
| General Purpose (GP_5_Gen5_1)  |              |
| Cost per GB (in USD)           | 0.13         |
| Max storage selected (in GB)   | x 41.6       |
| ESTIMATED STORAGE COST / MONTH | 5.26 USD     |
| COMPUTE COST / VCORE SECOND    | 0.000174 USD |

NOTES

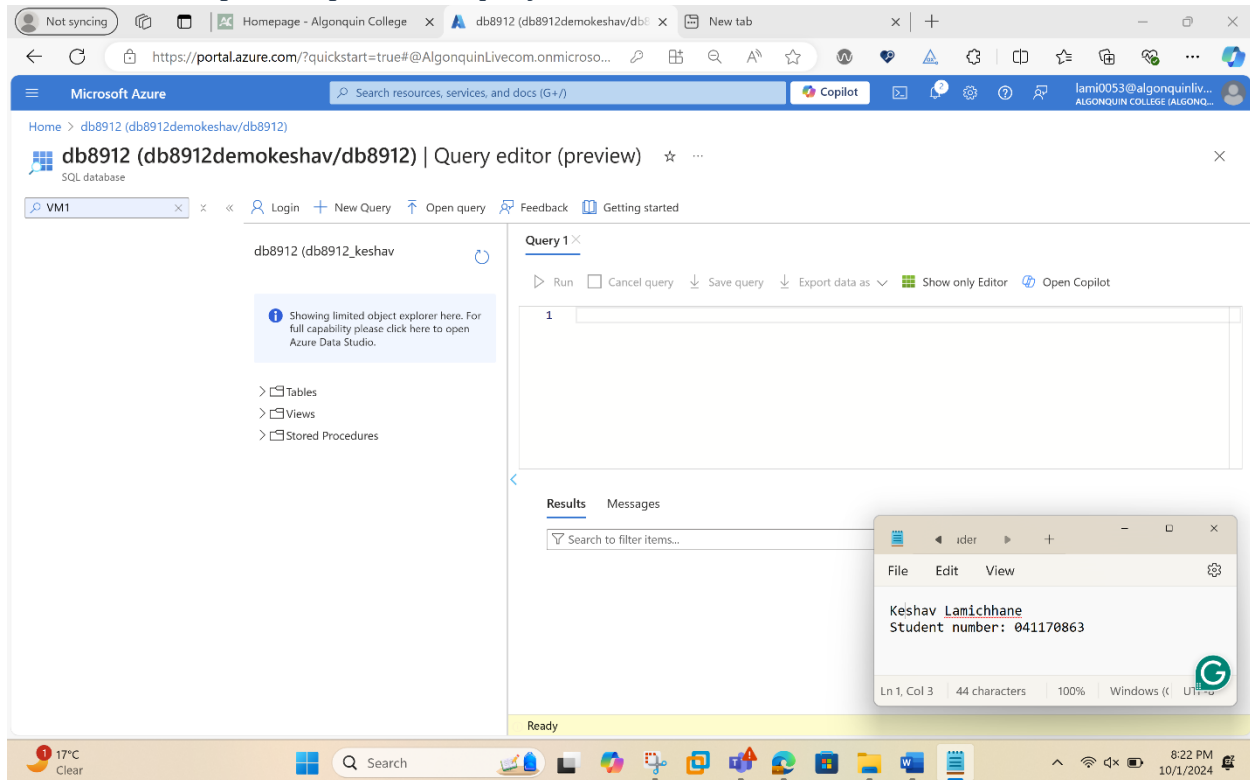
1 Serverless databases are billed in vCore seconds on a combination of CPU and memory utilisation, more about serverless billing

Review + create

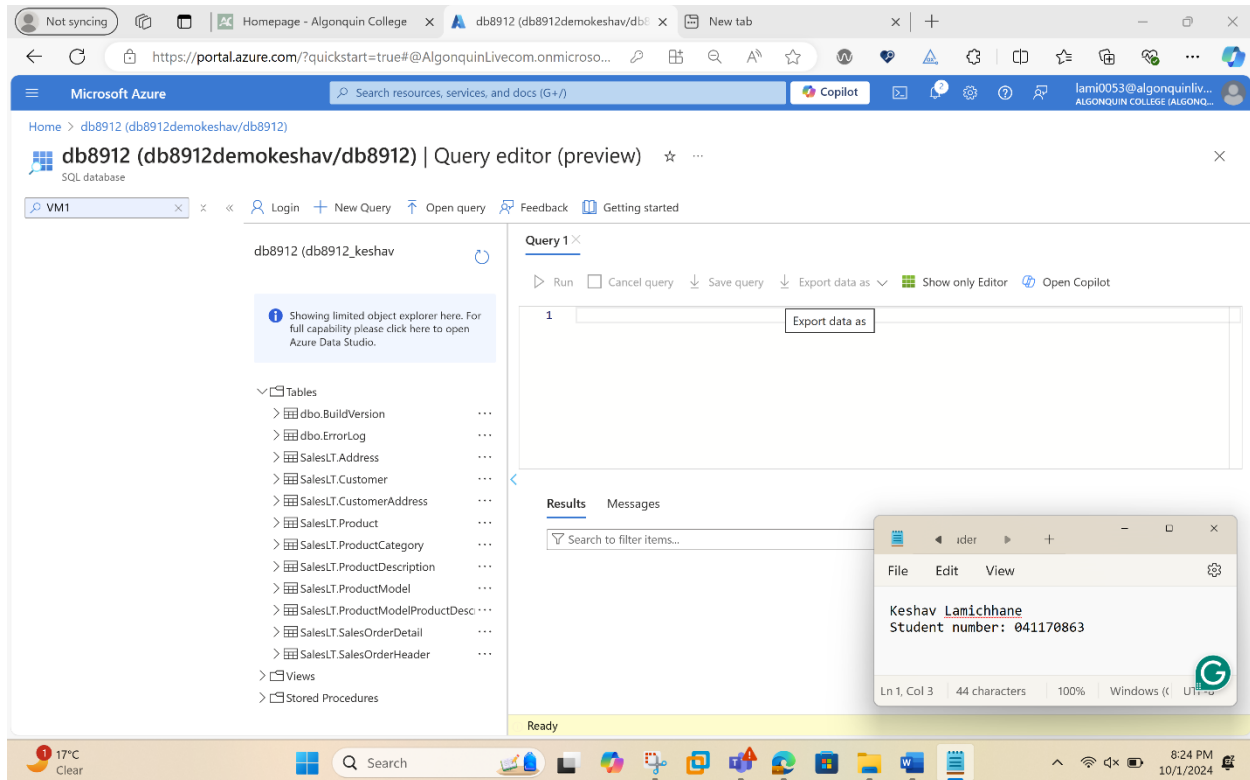
Step 7: On left side, click on Query editor (preview), and then sign in using the administrator login and password.



On the above step is completed, the query editor is shown below:



## Step 8: Expand the Tables folder to see the tables in the database



## Step 9: Try running the following queries in the query 1 pane; to do so, select Run.

```
SELECT ProductID, Name, ListPrice, ProductCategoryID
```

```
FROM SalesLT.Product;
```

```
SELECT p.ProductID, p.Name AS ProductName,  
       c.Name AS Category, p.ListPrice  
FROM SalesLT.Product AS p  
JOIN [SalesLT].[ProductCategory] AS c  
ON p.ProductCategoryID = c.ProductCategoryID;
```

db8912 (db8912demokeshav/db8912) | Query editor (preview)

SQL database

VM1

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Tables

- dbo.BuildVersion
- dbo.ErrorLog
- SalesLT.Address
- SalesLT.Customer
- SalesLT.CustomerAddress
- SalesLT.Product
- SalesLT.ProductCategory
- SalesLT.ProductDescription
- SalesLT.ProductModel
- SalesLT.ProductModelProductDesc
- SalesLT.SalesOrderDetail
- SalesLT.SalesOrderHeader
- Views
- Stored Procedures

Query 1

```

1 SELECT ProductID, Name, ListPrice, ProductCategoryID
2 FROM SalesLT.Product;
3
4 SELECT p.ProductID, p.Name AS ProductName,
5        c.Name AS Category, p.ListPrice
6 FROM SalesLT.Product AS p
7 JOIN [SalesLT].[ProductCategory] AS c
8     ON p.ProductCategoryID = c.ProductCategoryID;
9

```

Results Messages

Search to filter items...

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Ready

Click on run to get following screen

db8912 (db8912demokeshav/db8912) | Query editor (preview)

SQL database

VM1

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Tables

- dbo.BuildVersion
- dbo.ErrorLog
- SalesLT.Address
- SalesLT.Customer
- SalesLT.CustomerAddress
- SalesLT.Product
- SalesLT.ProductCategory
- SalesLT.ProductDescription
- SalesLT.ProductModel
- SalesLT.ProductModelProductDesc
- SalesLT.SalesOrderDetail
- SalesLT.SalesOrderHeader
- Views
- Stored Procedures

Query 1

```

1 SELECT ProductID, Name, ListPrice, ProductCategoryID
2 FROM SalesLT.Product;
3
4 SELECT p.ProductID, p.Name AS ProductName,
5        c.Name AS Category, p.ListPrice
6 FROM SalesLT.Product AS p
7 JOIN [SalesLT].[ProductCategory] AS c
8     ON p.ProductCategoryID = c.ProductCategoryID;
9

```

Results Messages

Search to filter items...

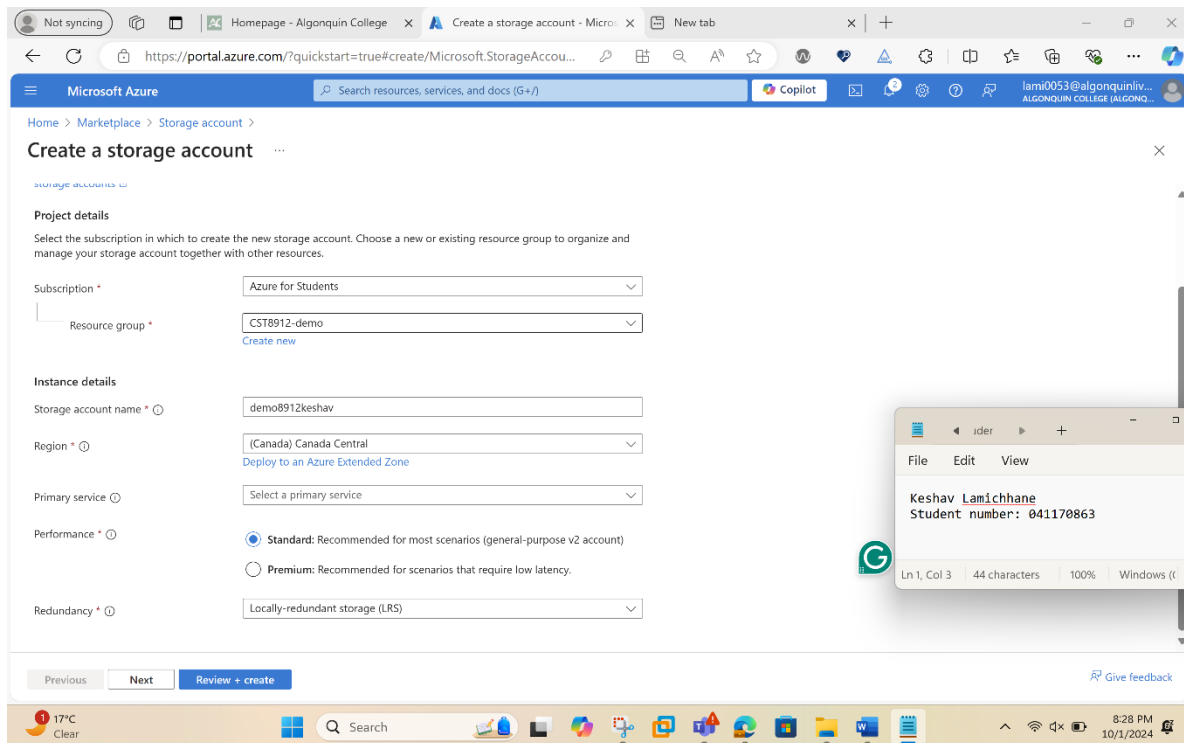
| ProductID | ProductName             | Category       | ListPrice |
|-----------|-------------------------|----------------|-----------|
| 771       | Mountain-100 Silver, 38 | Mountain Bikes | 3399.9900 |
| 772       | Mountain-100 Silver, 42 | Mountain Bikes | 3399.9900 |
| 773       | Mountain-100 Silver, 44 | Mountain Bikes | 3399.9900 |
| 774       | Mountain-100 Silver, 48 | Mountain Bikes | 3399.9900 |
| 775       | Mountain-100 Black, 38  | Mountain Bikes | 3374.0000 |

Query succeeded | 0s

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Student number: 041170863

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Step 10: Establish an Azure storage account using the following configurations, leaving the other advanced, networking, data security, and encryption settings at their default values.



**Project details**

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription \*

Resource group \*  [Create new](#)

**Instance details**

Storage account name \*

Region \*  [Deploy to an Azure Extended Zone](#)

Primary service

Performance \* ☒ **Standard:** Recommended for most scenarios (general-purpose v2 account)  
☐ **Premium:** Recommended for scenarios that require low latency.

Redundancy \*

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

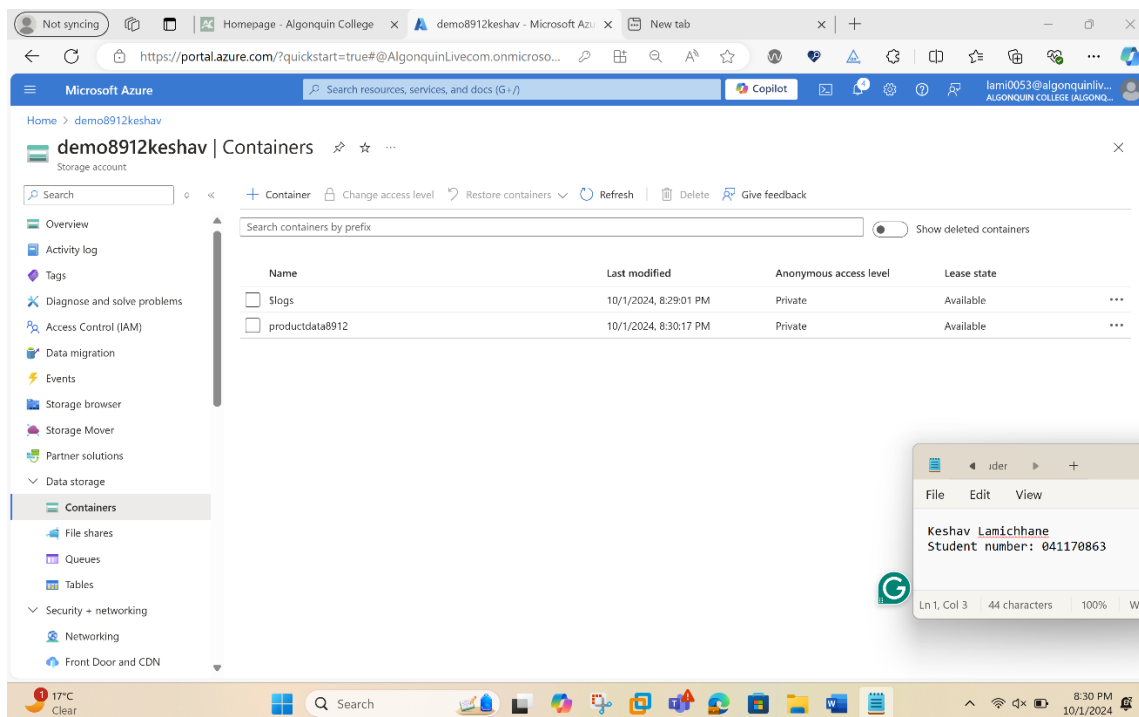
**Windows Taskbar:** 17°C Clear, Search, Taskbar icons, 8:28 PM 10/1/2024

**Floating Text Box:**

```

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Ln 1, Col 3 | 44 characters | 100% | Windows (t
  
```

Step 11: Now create a container “productdata8912” in storage account in this step.



**demo8912keshav | Containers**

Storage account

Search containers by prefix  ☐ Show deleted containers

| Name                                     | Last modified         | Anonymous access level | Lease state |
|------------------------------------------|-----------------------|------------------------|-------------|
| <input type="checkbox"/> \$logs          | 10/1/2024, 8:29:01 PM | Private                | Available   |
| <input type="checkbox"/> productdata8912 | 10/1/2024, 8:30:17 PM | Private                | Available   |

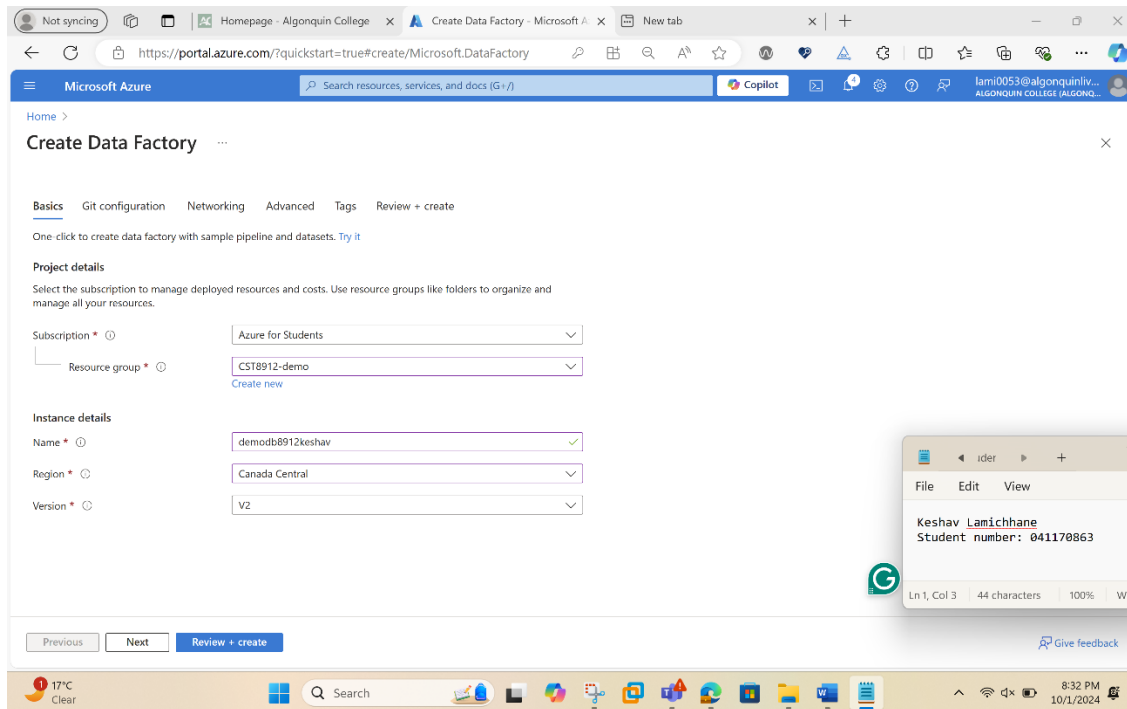
**Windows Taskbar:** 17°C Clear, Search, Taskbar icons, 8:30 PM 10/1/2024

**Floating Text Box:**

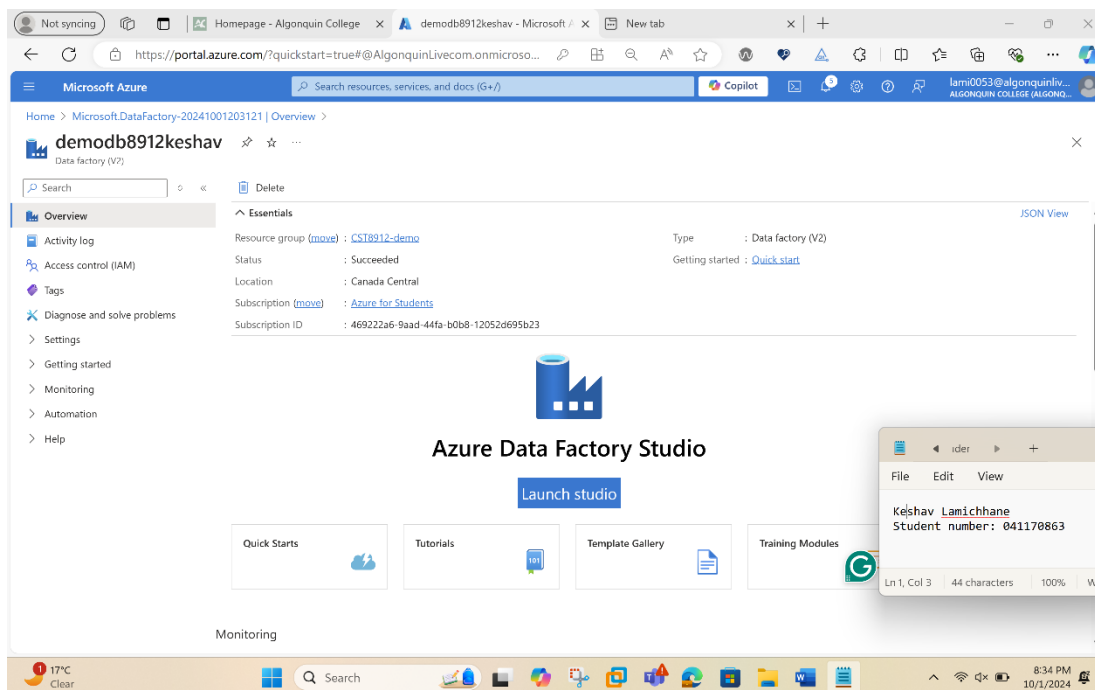
```

Keshav Lamichhane
Student number: 041170863
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```

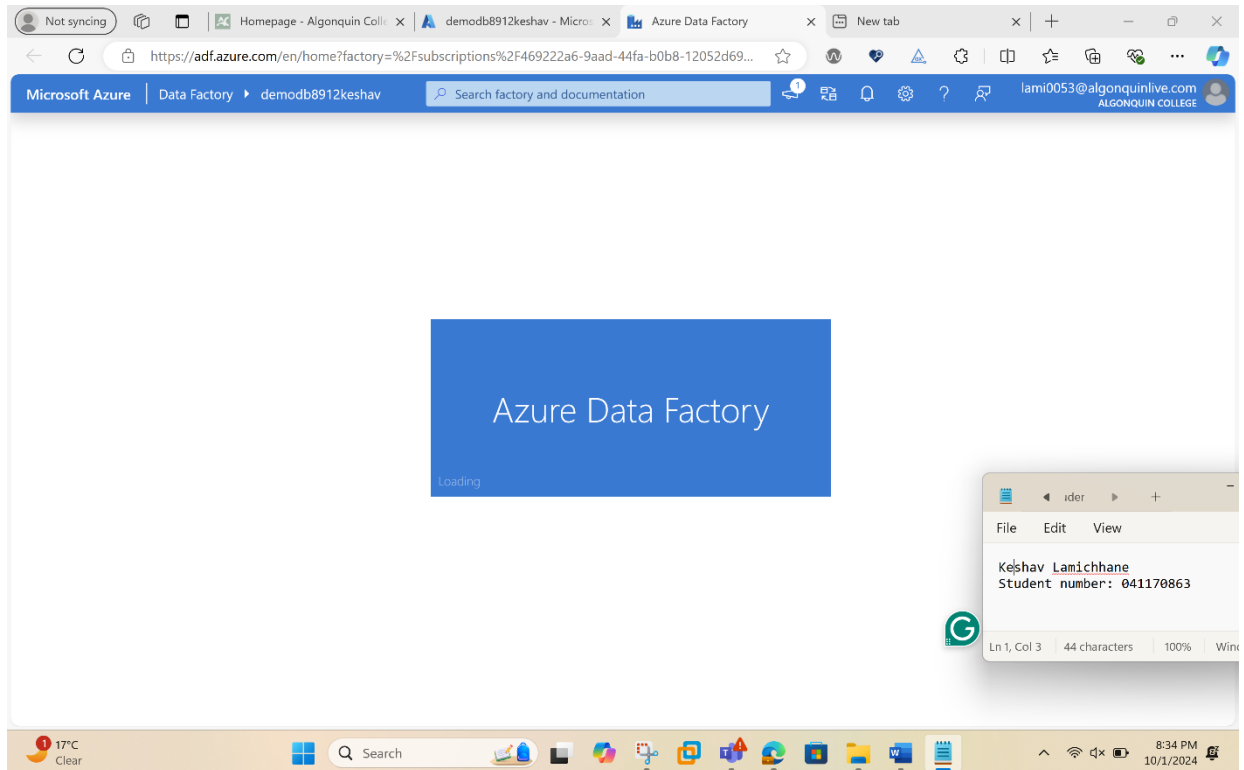
Step 12: Using the Azure portal, create a new resource in your resource group and search for Azure Data Factory. Leave Git configuration, networking, and advanced set to default.



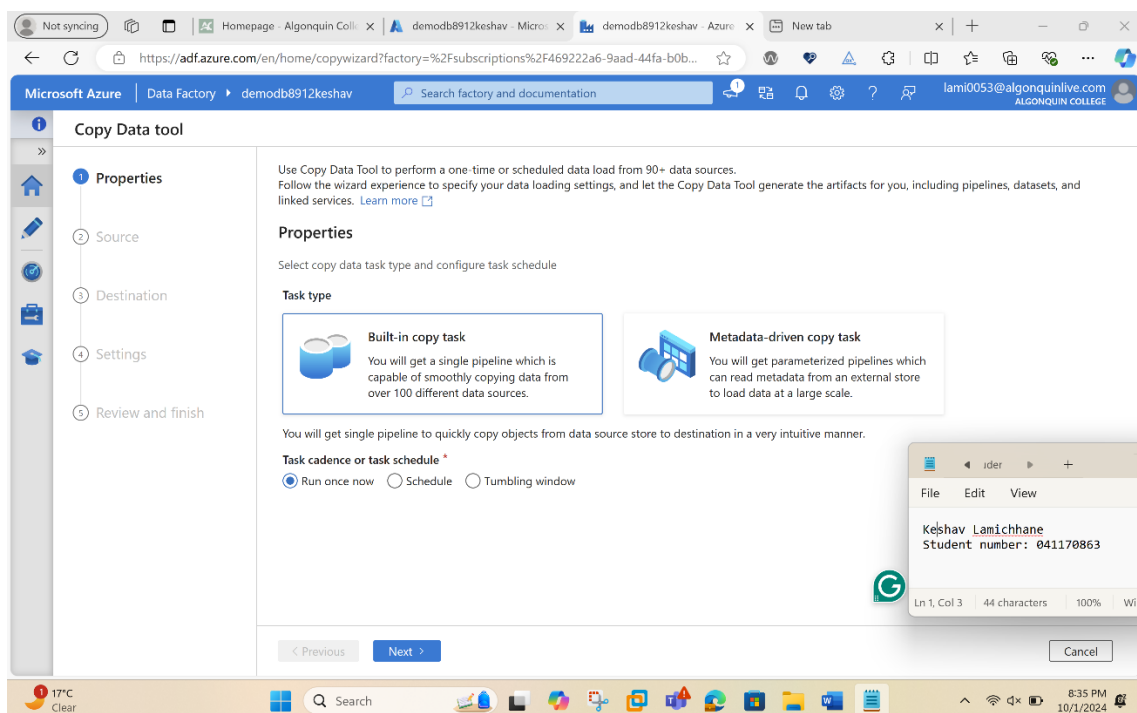
Step 13: Once created launch azure data factory studio.



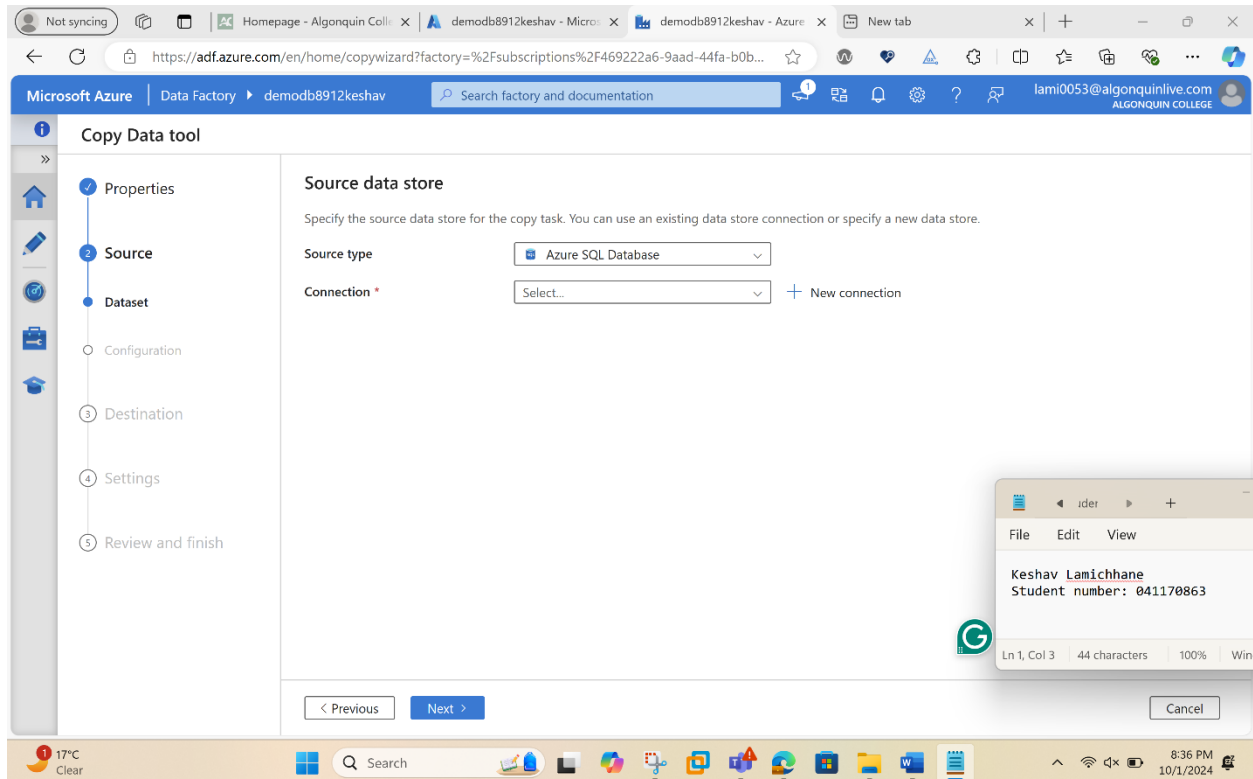
Step 14: On home page, click on launch data factory. choose option to ingest data. Then it will come up with following page:



Step 15: Choose task type, built in copy task, and task cadence “run once now”.



Step 16: Select the Azure SQL Database source type from the dropdown menu. For the connection, select a new one with the setup shown below, then verify the connection.



In this step, we create a new connection to azure sql database from data factory. Correct database name and password that we created while creating azure sql database is needed.

Microsoft Azure | Data Factory | demodb8912keshav

## Copy Data tool

Source data store

Specify the source data store for the copy task. You can use

Source type: Azure SQL Database

Connection: Select...

Version: Recommended (Legacy)

Import from connection string

Account selection method: From Azure subscription (Enter manually)

Azure subscription: Azure for Students (469222a6-9aad-44fa-b0b8-12052d695b23)

Server name: db8912demokeshav

Database name: db8912

Authentication type: SQL authentication

User name: db8912\_keshav

Password: Azure Key Vault

Create Cancel Test connection

Properties Source Dataset Configuration Destination Settings Review and finish

17°C Clear 8:42 PM 10/1/2024

Now we click on test connection to see following output successful connection.

Microsoft Azure | Data Factory | demodb8912keshav

## Copy Data tool

Source data store

Specify the source data store for the copy task. You can use

Source type: Azure SQL Database

Connection: Select...

Authentication type: SQL authentication

User name: db8912\_keshav

Password: Password Azure Key Vault

Always encrypted: ☐

Encrypt: Mandatory

Trust server certificate: ☐

Host name in certificate

Additional connection properties: New

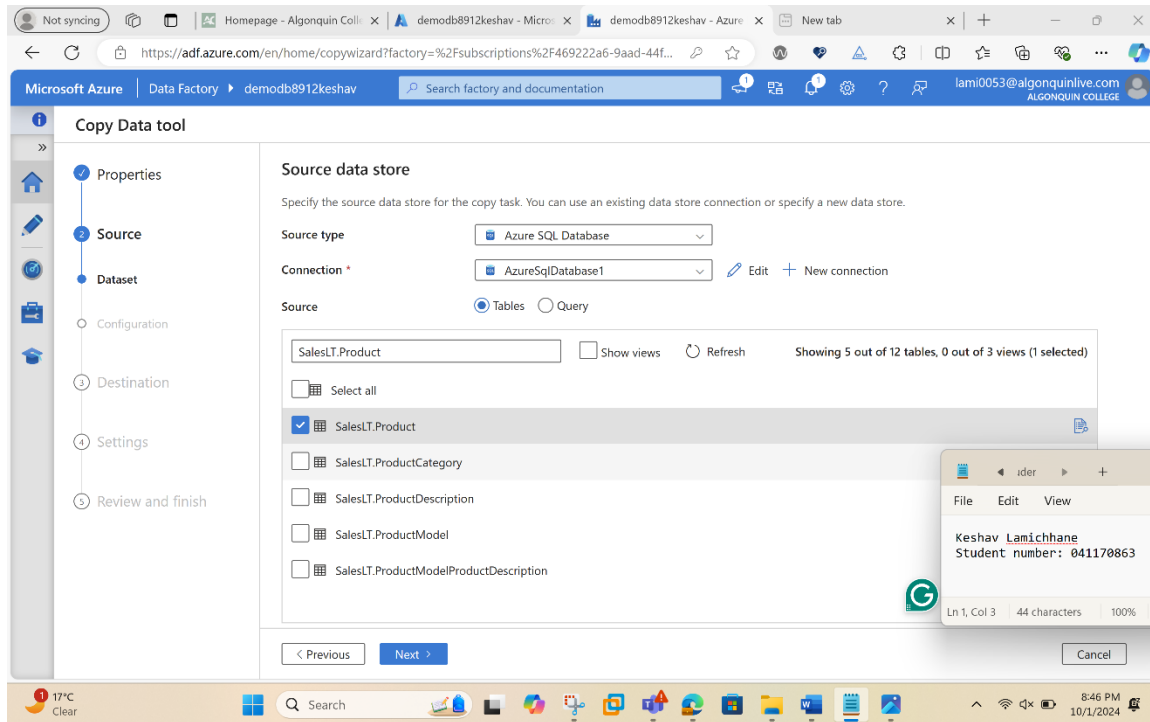
Create Cancel Test connection

Connection successful

Properties Source Dataset Configuration Destination Settings Review and finish

17°C Clear 8:42 PM 10/1/2024

Step 17: In this step, we select source table as “SalesLT.Product” from the dropdown, click next and you can preview data.



**Copy Data tool**

**Source data store**

Specify the source data store for the copy task. You can use an existing data store connection or specify a new data store.

Source type: Azure SQL Database

Connection: AzureSqlDatabase1

Source: Tables

SalesLT.Product

Select all

SalesLT.Product

SalesLT.ProductCategory

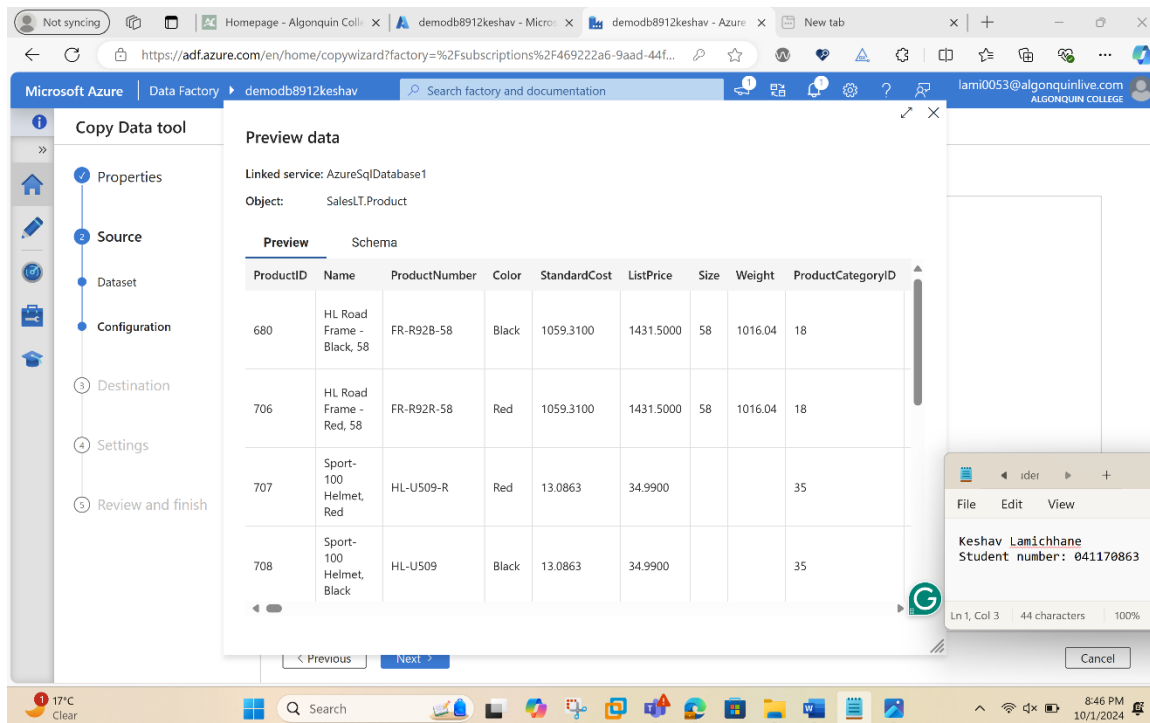
SalesLT.ProductDescription

SalesLT.ProductModel

SalesLT.ProductModelProductDescription

< Previous Next >

Next step is: click on preview.



**Copy Data tool**

**Preview data**

Linked service: AzureSqlDatabase1

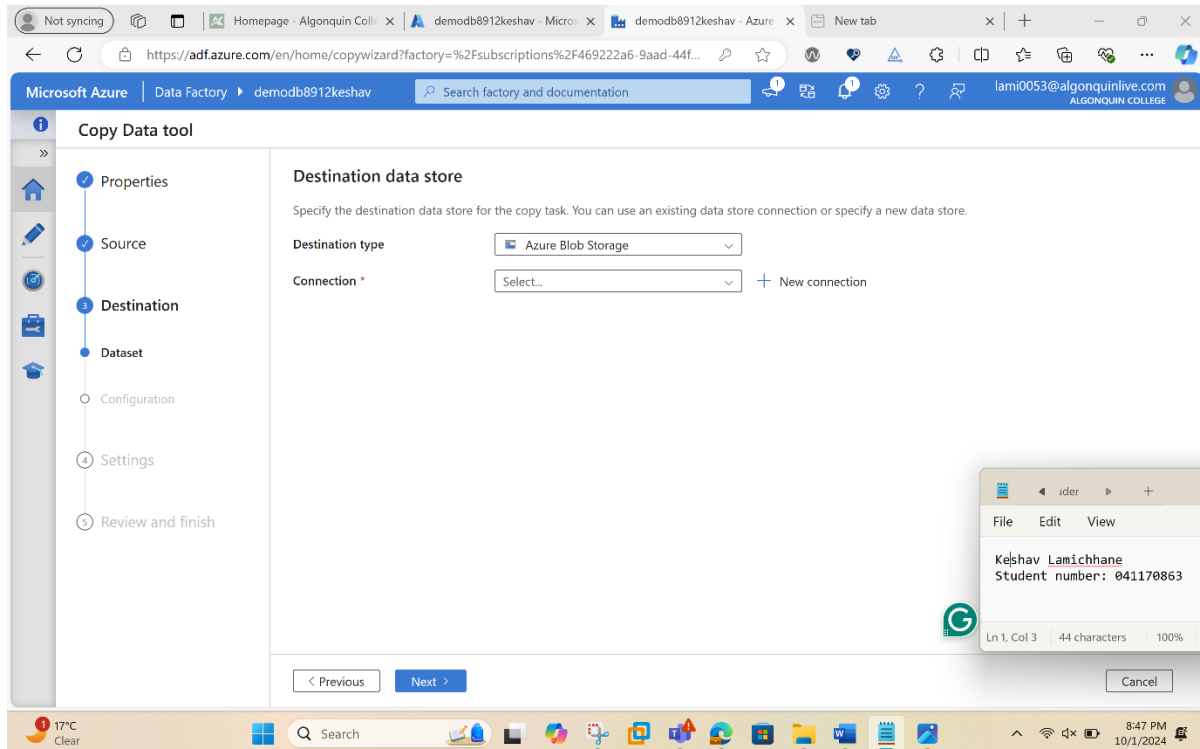
Object: SalesLT.Product

Preview Schema

| ProductID | Name                      | ProductNumber | Color | StandardCost | ListPrice | Size | Weight  | ProductCategoryID |
|-----------|---------------------------|---------------|-------|--------------|-----------|------|---------|-------------------|
| 680       | HL Road Frame - Black, 58 | FR-R92B-58    | Black | 1059.3100    | 1431.5000 | 58   | 1016.04 | 18                |
| 706       | HL Road Frame - Red, 58   | FR-R92R-58    | Red   | 1059.3100    | 1431.5000 | 58   | 1016.04 | 18                |
| 707       | Sport-100 Helmet, Red     | HL-U509-R     | Red   | 13.0863      | 34.9900   |      |         | 35                |
| 708       | Sport-100 Helmet, Black   | HL-U509       | Black | 13.0863      | 34.9900   |      |         | 35                |

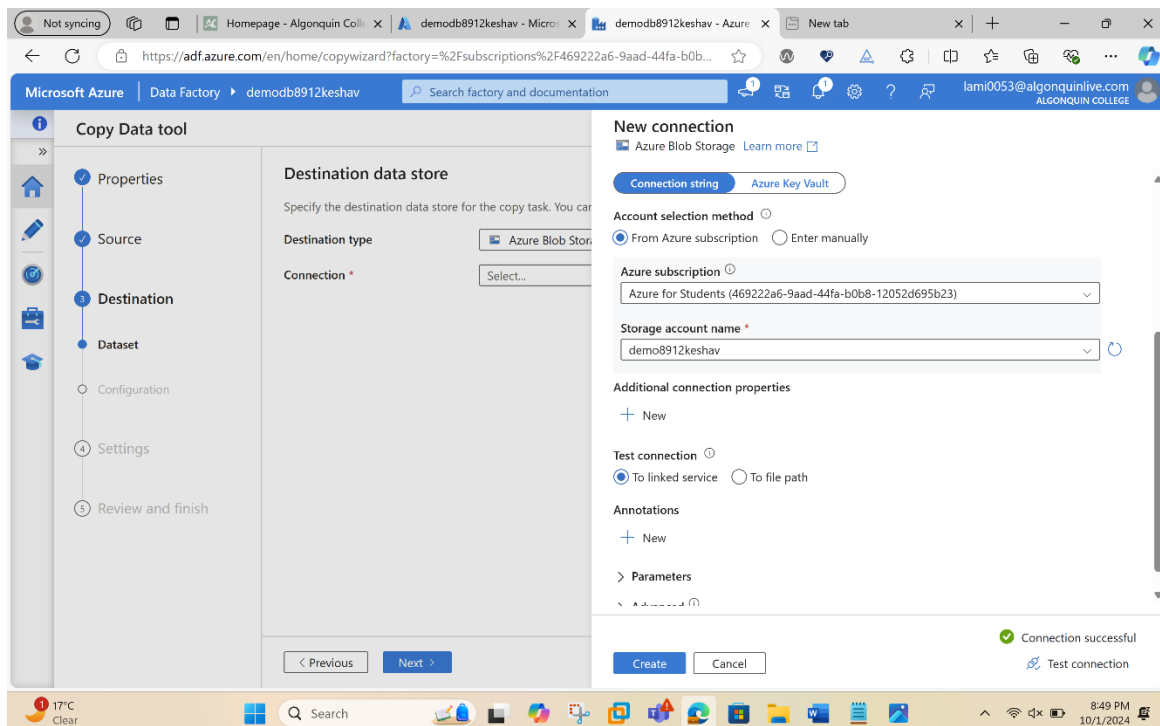
< Previous Next >

Step 18: In this step, we click next, choose destination type, select “Azure Blob Storage” from dropdown menu.



The screenshot shows the Microsoft Azure Data Factory Copy Data tool interface. The 'Destination data store' section is active, showing 'Destination type' set to 'Azure Blob Storage' and 'Connection' set to 'Select...'. A 'New connection' button is visible. A small text box overlay shows 'Kelshav Lamichhane' and 'Student number: 041170863'.

Step 19: Now we create new connection and test connection to storage account.



The screenshot shows the Microsoft Azure Data Factory Copy Data tool interface with the 'New connection' dialog open. The 'Account selection method' is set to 'From Azure subscription'. The 'Azure subscription' is 'Azure for Students (469222a6-9aad-b0b8-12052d695b23)'. The 'Storage account name' is 'demo8912keshav'. The 'Test connection' button is highlighted, and a green checkmark indicates 'Connection successful'.

Then we click on create.

The screenshot shows the 'Copy Data tool' configuration in the Microsoft Azure portal. The 'Destination' step is selected in the left-hand navigation pane. The main area displays the 'Destination data store' configuration. The 'Destination type' is set to 'Azure Blob Storage'. The 'Connection' is set to 'AzureBlobStorage1'. The 'Folder path' is 'productdata8912'. The 'File name' is 'product'. The 'Max concurrent connections' is set to 1. The 'Block size (MB)' is set to 1. The 'Metadata' section has a '+ New' button. A small text box in the bottom right corner of the screen displays 'Keshav Lamichhane' and 'Student number: 041170863'.

Microsoft Azure | Data Factory | demodb8912keshav

Copy Data tool

Properties

Source

Destination

Dataset

Configuration

Settings

Review and finish

Destination data store

Specify the destination data store for the copy task. You can use an existing data store connection or specify a new data store.

Destination type: Azure Blob Storage

Connection: AzureBlobStorage1

Folder path: productdata8912

File name: product

Max concurrent connections: 1

Block size (MB): 1

Metadata: + New

Keshav Lamichhane  
Student number: 041170863

Step 20: Choose the configuration as shown below:

The screenshot shows the 'Copy Data tool' configuration in the Microsoft Azure portal. The 'File format settings' step is selected in the left-hand navigation pane. The main area displays the 'File format settings' configuration. The 'File format' is set to 'DelimitedText'. The 'Column delimiter' is set to 'Comma (,)' and the 'Row delimiter' is set to 'Default (\r\n, or \n\r\n)'. The 'Add header to file' checkbox is checked. The 'Compression type' is set to 'Select...'. The 'Max rows per file' is set to 1. The 'File name prefix' is set to 'productdata8912'. A small text box in the bottom right corner of the screen displays 'Keshav Lamichhane' and 'Student number: 041170863'.

Microsoft Azure | Data Factory | demodb8912keshav

Copy Data tool

Properties

Source

Destination

Dataset

Configuration

Settings

Review and finish

File format settings

File format: DelimitedText

Column delimiter: Comma (,)

Row delimiter: Default (\r\n, or \n\r\n)

Add header to file: ☒

Advanced

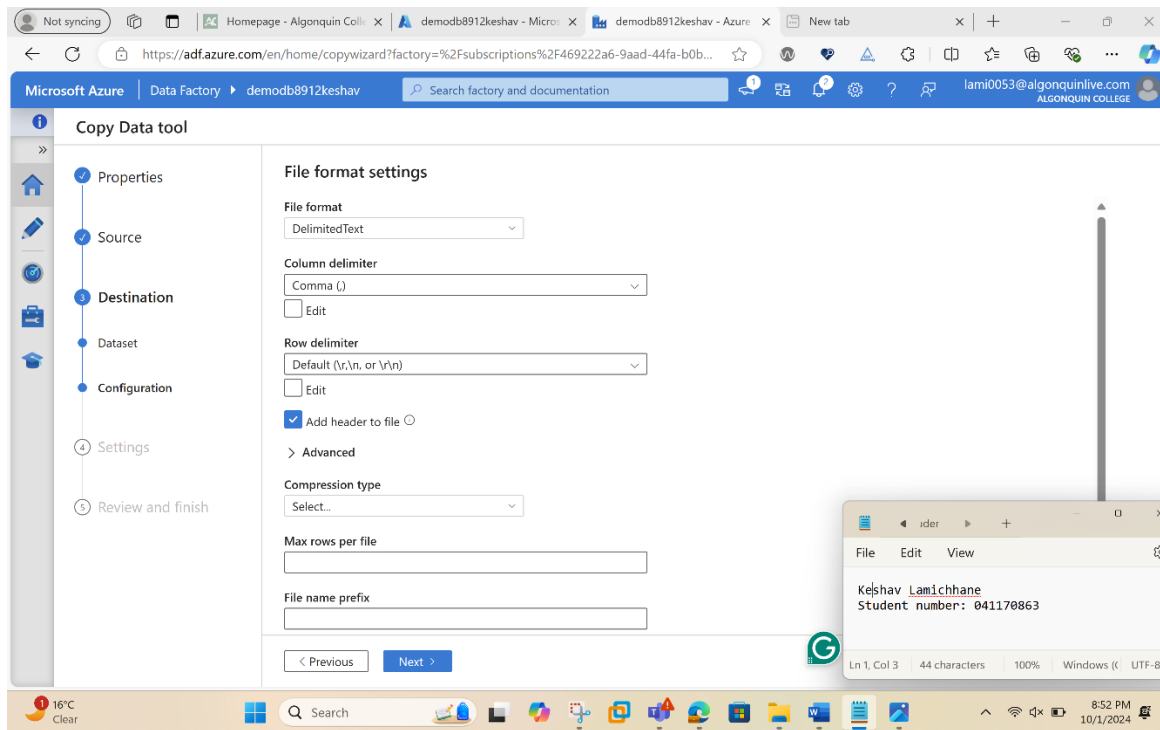
Compression type: Select...

Max rows per file: 1

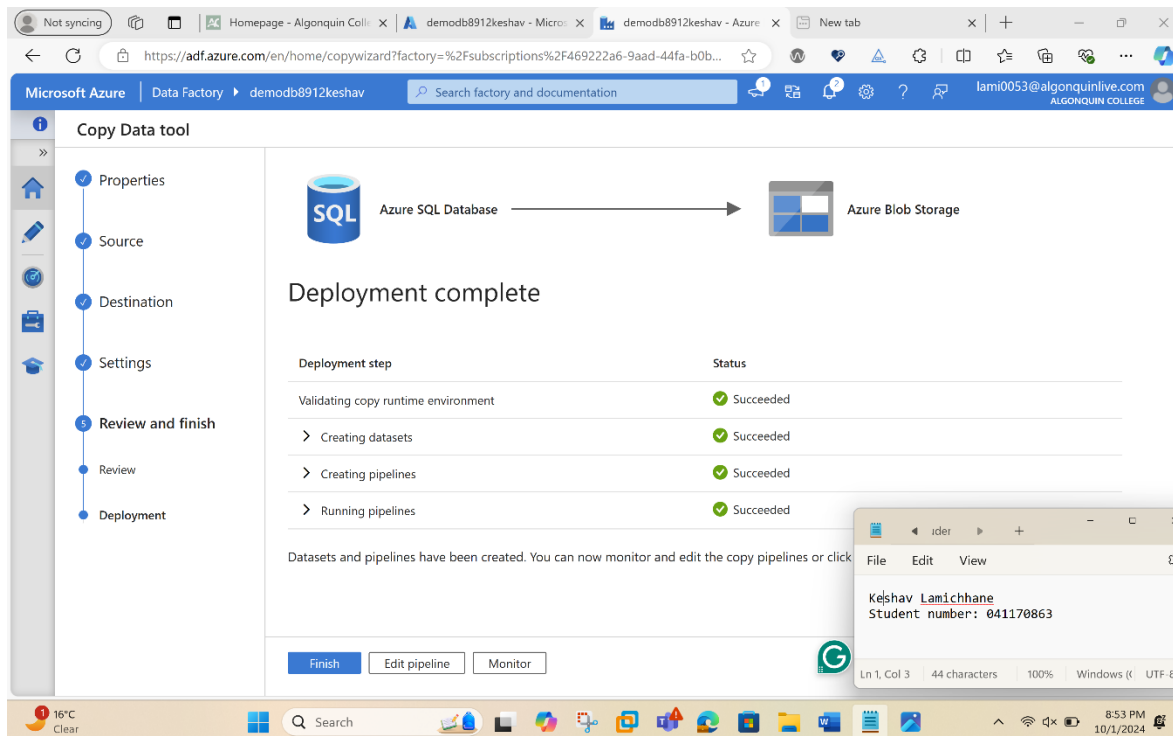
File name prefix: productdata8912

Keshav Lamichhane  
Student number: 041170863

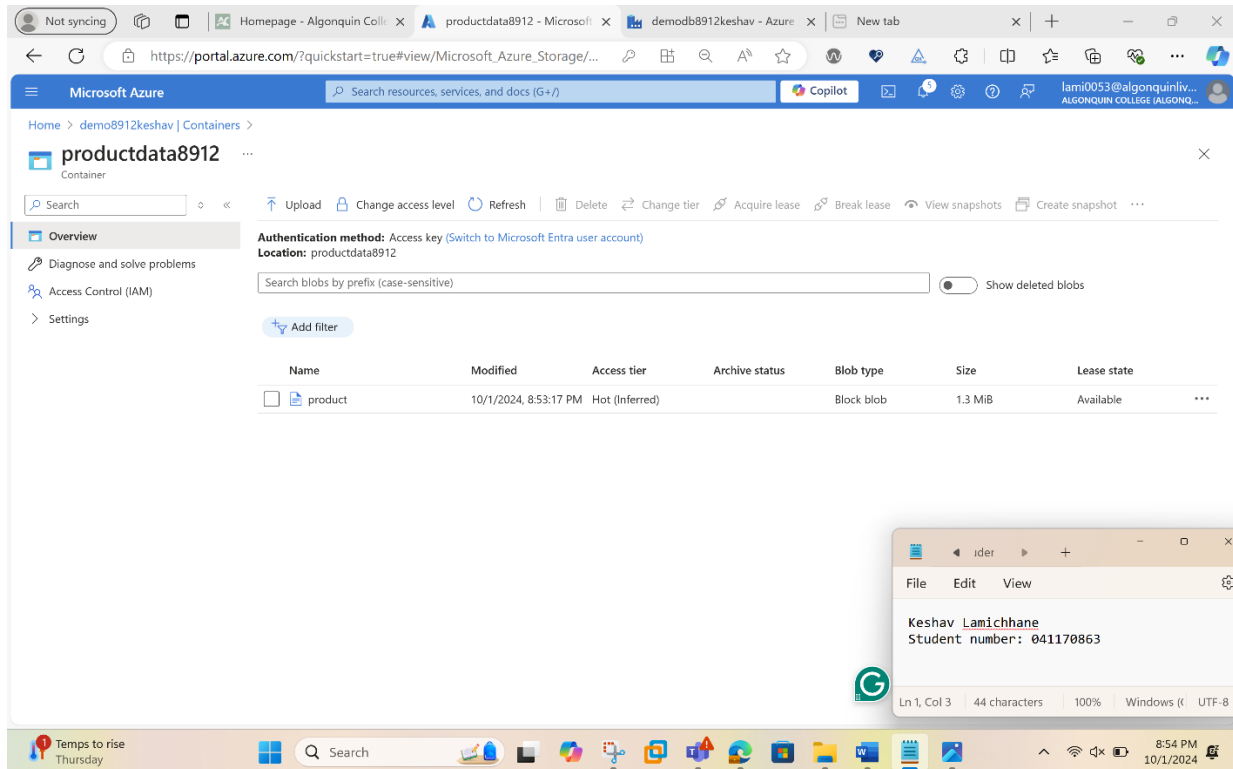
Then we click on Next.



Now leave the setting as default and click on next three times and we come with pipeline successful step.



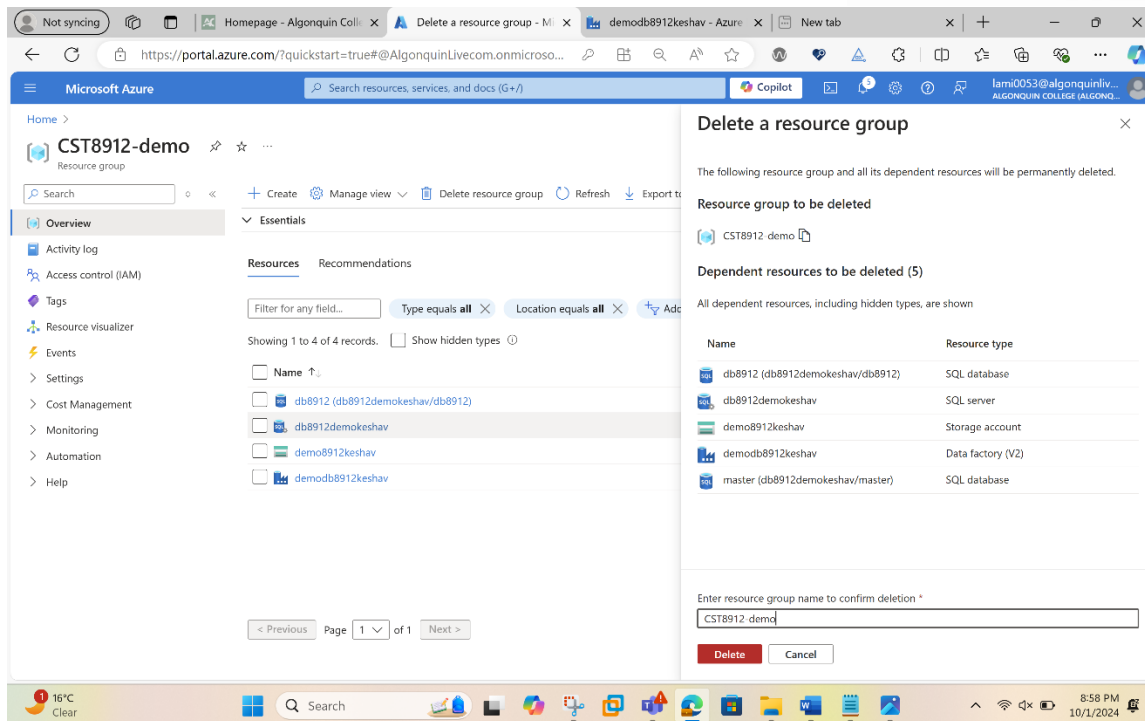
Step 21: This step is almost final step. In this step, we reviewed and finished this pipeline and checked the storage account container to see the product csv file has been copied from the database to storage account. The following step confirms the copying process.



The screenshot shows the Microsoft Azure portal interface. The main content area displays the 'productdata8912' container. The container's overview shows a single blob named 'product' with a size of 1.3 MiB, modified on 10/1/2024 at 8:53:17 PM. The authentication method is 'Access key' and the location is 'productdata8912'. A small window in the bottom right corner displays the user's name 'Keshav Lamichhane' and student number '041170863'.

| Name    | Modified              | Access tier    | Archive status | Blob type  | Size    | Lease state |
|---------|-----------------------|----------------|----------------|------------|---------|-------------|
| product | 10/1/2024, 8:53:17 PM | Hot (Inferred) |                | Block blob | 1.3 MiB | Available   |

Step 22: After completion of lab, we deleted all the resources created during this lab and finally documented the process.



This step completes the lab.

## Findings and analysis:

- To show how simple it is to install and manage databases in Azure, we built a SQL database utilizing a PaaS approach. Network setups were also addressed in the lab, including how to set up firewall rules to permit access from IP addresses.
- We executed basic SQL queries to retrieve and join data from the sample tables in the database, which are essential skills for working with relational databases.
- The lab used firewall settings and redundant backup options to emphasize the value of data storage and security. Microsoft Defender for SQL was not activated, however its availability is noted in case security is a major concern in the future.
- Deleting resources after completion helps reduce unwanted charges and keeps the cloud environment orderly.

## References:

*How to create an Azure SQL database | Azure Portal Series.* (n.d.). Wwww.youtube.com.

[https://www.youtube.com/watch?v=p7X8IH\\_XMtI](https://www.youtube.com/watch?v=p7X8IH_XMtI)

Microsoft Azure. (2019, July 31). *How to create an Azure SQL database using all configuration options | Azure Portal Series.* YouTube.

<https://www.youtube.com/watch?v=v54UCzJJAYY>

*Create a MySQL database on AWS!* (n.d.). Wwww.youtube.com.

<https://www.youtube.com/watch?v=v3jH1YxJqaY>